Supplemental material for "Taming multiparticle entanglement"

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APPENDIX A

In this appendix, we introduce an entanglement monotone for genuine multipartite entanglement that is motivated by the notion of fully decomposable entanglement witnesses. Let us point out that this defined quantity equals the negativity for the bipartite case [1], hence it can be considered as an extension of the negativity to the multipartite case.

Proposition 1. *For a generic multipartite state ϱ, consider*

$$N(\varrho) = -\min_{W\in\mathcal{W}} \mathrm{Tr}(\varrho W), \qquad (1)$$

$$\mathcal{W} = \Big\{ W \Big| \, \forall M : \exists\, 0 \leq P_M, Q_M \leq \mathbb{1} : W = P_M + Q_M^{T_M} \Big\},$$

where M stands for a possible partition of the subsystems. Then $N(\varrho)$ has the following properties:

- $N(\varrho^{\mathrm{bs}}) = 0$ *for all biseparable states* ϱ^{bs}.
- $N[\Lambda_{\mathrm{LOCC}}(\varrho)] \leq N(\varrho)$ *for all full LOCC operations.*
- $N(U_{\mathrm{loc}}\varrho U_{\mathrm{loc}}^\dagger) = N(\varrho)$ *for local basis changes* U_{loc}.
- $N(\sum_i p_i\varrho_i) \leq \sum_i p_i N(\varrho_i)$ *holds for all convex combinations* $\sum_i p_i\varrho_i$.

Proof. The first statement follows directly from the fact that any fully decomposable witness can only detect genuine multipartite entanglement, hence the expectation value satisfies $\mathrm{Tr}(\varrho^{\mathrm{bs}}W) \geq 0$ and vanishes for $W = 0$.

For the second property we effectively show $N[\Lambda(\varrho)] \leq N(\varrho)$ for all trace-preserving, completely positive operations $\Lambda(\varrho) = \sum_i K_i\varrho K_i^\dagger$ with $\sum_i K_i^\dagger K_i = \mathbb{1}$ that admit a fully separable operator-sum representation, which means that each operator $K_i = A_i \otimes B_i \otimes \cdots \otimes F_i$ has a tensor product form. This set of operations defines a superset to the set of full LOCC operations, so we verify the above property for an even larger set of possible operations [2]. Let us point out that for each operation Λ, there exists an adjoint operation $\Lambda^\dagger(Y) = \sum_i K_i^\dagger Y K_i$, that satisfies the identity $\mathrm{Tr}[\Lambda(X)Y] = \mathrm{Tr}[X\Lambda^\dagger(Y)]$ for all linear operators X, Y. The trace-preserving condition for Λ translates to a unital condition for the adjoint map $\Lambda^\dagger(\mathbb{1}) = \mathbb{1}$.

Let us first prove the following statement: For each trace-preserving, separable operation Λ and for any decomposable operator W the observable $\Lambda^\dagger(W)$ is decomposable as well. Suppose that $W = P + Q^{T_M}$ is an appropriate decomposition [3] with respect to a chosen partition M. Because of linearity we obtain $\Lambda^\dagger(W) = \Lambda^\dagger(P) + \Lambda^\dagger(Q^{T_M})$. First, we want to check "normalization" $0 \leq \Lambda^\dagger(P) \leq \mathbb{1}$. Complete positivity provides $\Lambda^\dagger(P) \geq 0$ since $P \geq 0$ is positive semidefinite itself. If one applies the adjoint map to $(\mathbb{1} - P) \geq 0$ and employs the unital condition one obtains

$$\Lambda^\dagger(\mathbb{1} - P) = \Lambda^\dagger(\mathbb{1}) - \Lambda^\dagger(P) = \mathbb{1} - \Lambda^\dagger(P) \geq 0, \qquad (2)$$

hence the upper bound $\Lambda^\dagger(P) \leq \mathbb{1}$ holds as well. We can apply the same argument to $\Lambda^\dagger(Q)$ if we can fulfil the identity $\Lambda^\dagger(Q^{T_M}) = [\tilde\Lambda^\dagger(Q)]^{T_M}$ with $\tilde\Lambda$ being completely positive and unital. Using the assumed tensor product structure of each operator K_i it is straightforward to deduce the Kraus operators of the liner map $\tilde\Lambda$ satisfying this identity. These operators $\tilde K_i = \tilde A_i \otimes \tilde B_i \otimes \cdots \otimes \tilde F_i$ are given by $\tilde A_i = A_i$ if $A \notin M$ and $\tilde A_i = A_i^*$ if $A \in M$, and similar relations for all other operators. Let us point out that this is the only step where one explicitly needs the separable operator structure.

Via this statement one finally obtains

$$\begin{aligned} N[\Lambda(\varrho)] &= -\min_{W\in\mathcal{W}} \mathrm{Tr}[\Lambda(\varrho)W] = -\min_{W\in\mathcal{W}} \mathrm{Tr}[\varrho\Lambda^\dagger(W)] \\ &\leq -\min_{W\in\mathcal{W}} \mathrm{Tr}[\varrho W] = N[\varrho]. \end{aligned} \qquad (3)$$

The inequality holds since $\Lambda^\dagger(W)$ is decomposable again, hence the optimization over the complete set of decomposable entanglement witnesses can only produce a lower (negative) expectation value.

Invariance under local basis changes is a direct consequence of the previous property. Since any local basis change U_{loc} represents an invertible full LOCC operation, one obtains

$$N(\varrho) \geq N(U_{\mathrm{loc}}\varrho U_{\mathrm{loc}}^\dagger) \geq N[U_{\mathrm{loc}}^\dagger(U_{\mathrm{loc}}\varrho U_{\mathrm{loc}}^\dagger)U_{\mathrm{loc}}] = N(\varrho).$$

Thus a local basis does not change the value of $N(\varrho)$.

The convexity statement

$$\begin{aligned} N\left(\sum_i p_i\varrho_i\right) &= -\min_W \sum_i p_i \mathrm{Tr}(\varrho_i W) \\ &\leq \sum_i p_i \left[-\min_W \mathrm{Tr}(\varrho_i W)\right] \\ &= \sum_i p_i N(\varrho_i), \end{aligned} \qquad (4)$$

follows from linearity of the trace and the fact that if one performs independent optimizations then the obtained expectation value can only be smaller. This completes the proof. □

APPENDIX B

In this appendix we prove that our criterion is optimal for three-qubit GHZ and W states and for four-qubit GHZ and cluster states (cf. marked states of Table I).

First, for the GHZ states mixed with white noise, the noise thresholds at which the states become biseparable were already derived in Ref. [4] and the values of p_{tol} for our criterion coincide with these values.

Second, for the three-qubit W state $|W_3\rangle$ we introduce a variation of our relaxation idea, in order to show that for a white noise contribution of $p \approx 0.521$ the corresponding state is biseparable.

Consider the tripartite case. Instead of supersets to approximate separable states for bipartition $A|BC$, we now consider a strictly smaller set of states, i.e., any state of this subset must necessarily be separable. States of this restricted class are denoted as $\tilde{\varrho}_{A|BC}$, and similar for other bipartitions. If a given state has a decomposition

$$\varrho = p_1 \tilde{\varrho}_{A|BC} + p_2 \tilde{\varrho}_{B|AC} + p_3 \tilde{\varrho}_{C|AB}, \tag{5}$$

then it is clearly biseparable. The main difficulty is to find operational subset approximations, though results have been obtained along this direction recently [5–7]. For the three-qubit case we employ an approximation that exploits the low dimensionality of the problem. Instead of all separable states of a $2 \otimes 4$ system we consider only separable states $\tilde{\varrho}_{A|BC}$ that are supported on the symmetric subspace of system BC. Since $\mathrm{Sym}(\mathcal{H}_{BC}) \simeq \mathbb{C}^3$, one effectively has a $2 \otimes 3$ system for which the PPT condition $\tilde{\varrho}_{A|BC}^{T_A} \geq 0$ becomes necessary and sufficient for separability [8]. Similar definitions are used for the other bipartitions. The search for an explicit decomposition as given by Eq. (5) can be cast into the form of a SDP again.

The solution for $\varrho(p) = (1-p)|W_3\rangle\langle W_3| + p\mathbb{1}/8$ is as follows: The separable states for $B|AC$ and $C|AB$ are the same as for the bipartition $A|BC$ but with appropriate permutations of the subsystems. All probabilities equal $p_i = 1/3$. Thus, it is left to characterize $\tilde{\varrho}_{A|BC}$. The corresponding eigenspectrum is given by $\lambda_1 = \lambda_2 = p/8$, $\lambda_3 = 3p/8$ and $\lambda_4 = 1 - 5p/8$ with respect to the eigenstates $|\chi_1\rangle = |000\rangle$, $|\chi_2\rangle = |111\rangle$,

$$|\chi_3\rangle = \left(2\sqrt{2}|1\rangle \otimes |\psi^+\rangle - |011\rangle\right)/3, \tag{6}$$

$$|\chi_4\rangle = a|0\rangle \otimes |\psi^+\rangle + \sqrt{1-a^2}|100\rangle, \tag{7}$$

with the abbreviation $|\psi^+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}$. The parameter $a \in \mathbb{R}$ is determined by the linear condition given by Eq. (5) and will be the root of a non-linear function. The only remaining condition is $\tilde{\varrho}_{A|BC}^{T_A} \geq 0$ which is satisfied if $p \geq \bar{p}$ with a critical white noise level given by

$$\bar{p} = \frac{1}{382}\left[367 - 71\sqrt{3} - \sqrt{2894\sqrt{3} - 2988}\right] \approx 0.52102. \tag{8}$$

At this white noise level one has $a \approx 0.98716$. The noise level $\bar{p}$ coincides with the maximal p_{tol} for our criterion.

Finally, for the four-qubit cluster state $|Cl_4\rangle$ we used the algorithm of Ref. [9], that can be used to prove different forms of multipartite separability. We found that a state with a noise contribution of $p = 0.616$ is biseparable, while the witness W_{Cl4} of Eq. (6) in the main text detects it for $p < 8/13 \approx 0.6154$ as genuine multipartite entangled.

APPENDIX C

The fully decomposable witness for the Dicke state $|D_{2,4}\rangle$ which was used in the main text to demonstrate entanglement detection by measuring a restricted set of observables is given by

$$W_D = \frac{1}{16}\left[\mathbb{1} + \alpha_1 X^{\otimes 4} + \alpha_1 Y^{\otimes 4} + \alpha_2 Z^{\otimes 4} + \alpha_3 \left(X_1X_2Y_3Y_4 + \mathrm{perms}\right) + \alpha_4 \left(Z_1Z_2Y_3Y_4 + \mathrm{perms}\right)\right. \\ \left. + \alpha_4 \left(Z_1Z_2X_3X_4 + \mathrm{perms}\right) + \alpha_5 \left(X_1X_2\mathbb{1}_3\mathbb{1}_4 + \mathrm{perms}\right) + \alpha_5 \left(Y_1Y_2\mathbb{1}_3\mathbb{1}_4 + \mathrm{perms}\right) + \alpha_6 \left(Z_1Z_2\mathbb{1}_3\mathbb{1}_4 + \mathrm{perms}\right)\right]. \tag{9}$$

Here, $X_1X_2Y_3Y_4 + \mathrm{perms}$ is the sum over all distinct permutations of $X_1X_2Y_3Y_4$. Moreover, $\alpha_1 = 0.014$, $\alpha_2 = -0.095$, $\alpha_3 = 0.0046$, $\alpha_4 = 0.16$, $\alpha_5 = -0.14$, $\alpha_6 = -0.15$.

APPENDIX D

Here, we prove the generalized n-qubit version W_{Cln} of the cluster state witnesses given in the main text to be fully decomposable witnesses. For four and seven qubits, W_{Cln} reduces to forms of Eqs. (6) and (8) of the main text, respectively. First, let us briefly introduce the no-

tation that we use.
Consider an arbitrary graph $G = (V, E)$ that is defined by a set V of vertices which correspond to qubits and a set E of edges that connect some of these vertices.
Then, by $\mathcal{N}(i)$ we denote the *neighborhood of qubit* i, i.e., the set of all vertices that are connected to i by an edge. Moreover, we define $\widetilde{\mathcal{N}}(i) = \mathcal{N}(i) \cup \{i\}$.
The commuting operators defined by

$$g_i = X_i \prod_{k\in\mathcal{N}(i)} Z_k,\ i = 1, \ldots, n \tag{10}$$

are the generators of the so-called *stabilizer group*, i.e., $\{S_1, \ldots, S_{2^n}\} = \langle g_1, \ldots, g_n\rangle$. In other words, every stabilizer S_i can be written as a product of some (or all) of these generators g_i.
The *graph state* $|G\rangle$ associated to the given graph G is then uniquely defined by

$$g_i|G\rangle = |G\rangle,\ \forall\, i = 1, \ldots, n\,. \tag{11}$$

One can also associate an orthonormal basis to a graph G. The elements $|a_1 \ldots a_n\rangle_G$ of this so-called *graph state basis* are defined by

$$g_i|a_1 \ldots a_n\rangle_G = (-1)^{a_i}|a_1 \ldots a_n\rangle_G,\ \forall\, i = 1, \ldots, n\,. \tag{12}$$

Consequently, $|G\rangle = |0 \ldots 0\rangle_G$. Also, note that projectors on these vectors can be written as

$$_G|a_1 \ldots a_n\rangle\langle a_1 \ldots a_n|_G = \prod_{i=1}^{n} \frac{(-1)^{a_i} g_i + \mathbb{1}}{2}\,. \tag{13}$$

For better readability, we will drop the subscript G in the following and write $|a_1 \ldots a_n\rangle_G = |\vec{a}\rangle$. Thus, except for a single exception that we will explicitly mention, all vectors in the following have to be understood in the graph state basis. However, it is important to note that all partial transpositions will be done w.r.t the computational basis.
A *partition* M is a subset of the set of all qubits, while we refer to the set made up of partition M and its complement $\overline{M}$ as *bipartition* $M|\overline{M}$.
With the notation clarified, we first prove three lemmas to prepare the proof that the operator W_{Cl} presented in the paper is a fully decomposable witness. The first of these lemmas starts from two orthogonal graph state basis vectors and shows which kind of partial transpositions one can apply to one of these vectors such that the two resulting vectors are orthogonal. Lemma 2 supplies an upper bound on the eigenvalues of any partially transposed state in terms of the state's Schmidt coefficients. Finally, Lemma 3 provides some kind of construction manual. Given a qubit and starting from a graph state basis vector, it tells us how we can construct expressions out of this basis vector which are invariant under a partial transposition on the given qubit.
These lemmas will then be used in Proposition 2 to prove that the presented operator W_{Cln} is a fully decomposable witness.

Lemma 1. *Given a graph $G = (V, E)$ of n qubits and an arbitrary bipartition $M|\overline{M}$ of these qubits. Let $|\vec{a}\rangle$ and $|\vec{b}\rangle$ be two arbitrary states in the associated graph state basis. Let $\mathcal{U}_M$ be the set of all qubits that lie in the same partition as their neighbors, i.e., $\mathcal{U}_M = \{i \,|\, \widetilde{\mathcal{N}}(i) \subseteq M \vee \widetilde{\mathcal{N}}(i) \subseteq \overline{M}\}$. If there is a qubit $i \in \mathcal{U}_M$, s.t. $b_i \neq a_i$, then*

$$\langle\vec{b}|\,(|\vec{a}\rangle\langle\vec{a}|)^{T_M}\,|\vec{b}\rangle = 0 \tag{14}$$

Proof. Let $g_i, i = 1 \ldots n$ be the generators defined by Eq. (10). Since $X^T = X$, $Y^T = -Y$, $Z^T = Z$ and $\mathbb{1}^T = \mathbb{1}$, the partial transposition of a product of generators only changes the product's sign. Thus, we can describe the action of the partial transpose T_M w.r.t partition M on products of generators by

$$\left(\prod_{i=1}^{n} g_i^{x_i}\right)^{T_M} = (-1)^{f(\vec{x})}\left(\prod_{i=1}^{n} g_i^{x_i}\right)\,, \tag{15}$$

where $x_i \in \{0, 1\}$. Here, f depends on M and is a Boolean function defined by

$$f : \{0,1\}^n \rightarrow \{0,1\} \tag{16}$$
$$\vec{x} \mapsto f(\vec{x}) = \begin{cases} 0, & \text{if } \left(\prod_{i=1}^{n} g_i^{x_i}\right)^{T_M} = \prod_{i=1}^{n} g_i^{x_i} \\ 1, & \text{if } \left(\prod_{i=1}^{n} g_i^{x_i}\right)^{T_M} = -\prod_{i=1}^{n} g_i^{x_i} \end{cases}.$$

Note that the *support* $\mathrm{supp}(f)$ of a Boolean function contains the bits that the function depends on, i.e.,

$$\mathrm{supp}(f) = \{i \,|\, \exists\, \vec{x}, \text{s.t.} f(x_1, \ldots, x_i, \ldots, x_n) \neq f(x_1, \ldots, x_i \oplus 1, \ldots, x_n)\}\,. \tag{17}$$

Due to the explicit form of the g_i, flipping the value of x_i cannot change $f(\vec{x})$, if $i \in \mathcal{U}_M$. Therefore, $\mathcal{U}_M \cap \mathrm{supp}(f) = \{\}$.

For this reason, we can pull qubits in $\mathcal{U}_M$ out of the partial transposition T_M in the following way (using also Eq. (13)).

$$\langle\vec{b}|\left(|\vec{a}\rangle\langle\vec{a}|\right)^{T_M}|\vec{b}\rangle = \operatorname{Tr}\left[\prod_{j=1}^{n}\frac{(-1)^{b_j}g_j+\mathbb{1}}{2}\prod_{i\in\mathcal{U}_M}\frac{(-1)^{a_i}g_i+\mathbb{1}}{2}\left(\prod_{i\notin\mathcal{U}_M}\frac{(-1)^{a_i}g_i+\mathbb{1}}{2}\right)^{T_M}\right] \tag{18}$$

Since $\frac{g_i+\mathbb{1}}{2}\frac{-g_i+\mathbb{1}}{2}=0$, the last expression vanishes if there is an $i\in\mathcal{U}_M$, such that $b_i\neq a_i$. □

Lemma 2. *Given a state $|\psi\rangle$ and its Schmidt decomposition $|\psi\rangle=\sum_{i=1}^{d_1}\lambda_i|\mu_i\rangle\otimes|\nu_i\rangle$ with respect to some bipartition $M|\overline{M}$, where $\lambda_i\geq 0$, $d_1=\dim(M)$, $d_2=\dim(\overline{M})$ and w.l.o.g. $d_1\leq d_2$. Note that, here, $|\mu_i\rangle\otimes|\nu_i\rangle$ is not to be understood in a graph state basis. Then, for any state $|\phi\rangle$,*

$$\langle\phi|\left(|\psi\rangle\langle\psi|\right)^{T_M}|\phi\rangle\leq\max_i\lambda_i^2 \tag{19}$$

Proof. Writing down $(|\psi\rangle\langle\psi|)^{T_M}$ in the basis $\{|\mu_i\rangle\otimes|\nu_i\rangle\}_{i=1\ldots d_1,j=1\ldots d_2}$, one obtains a matrix with two different kinds of submatrices. First, a diagonal one with diagonal elements λ_i^2 or zero. Second, anti-diagonal submatrices of the form

$$\begin{pmatrix}0 & \lambda_i\lambda_j\\ \lambda_i\lambda_j & 0\end{pmatrix}. \tag{20}$$

Thus, the eigenvalues of the total matrix are $\{\pm\lambda_i\lambda_j,\lambda_i^2,0\}$ and the maximum of these eigenvalues has the form λ_i^2.

□

For the following lemma, we need to keep in mind that the application of the Pauli operator Z_k to a graph state basis vector creates a bit flip on bit k, i.e.,

$$Z_k|\vec{a}\rangle=|a_1\ldots a_{k-1}\,a_k\oplus 1\,a_{k+1}\ldots a_n\rangle\,. \tag{21}$$

Lemma 3. *Given a graph G. Then, in the associated graph state basis,*

$$\left(|\vec{a}\rangle\langle\vec{a}|+|\vec{b}\rangle\langle\vec{b}|\right)^{T_k}=|\vec{a}\rangle\langle\vec{a}|+|\vec{b}\rangle\langle\vec{b}|\,, \tag{22}$$

if $|\vec{b}\rangle$ is obtained from $|\vec{a}\rangle$ in one of the two following ways.

(i) $|\vec{b}\rangle=Z_k|\vec{a}\rangle$.

(ii) $|\vec{b}\rangle=\prod_{i\in\mathcal{N}(k)}Z_i|\vec{a}\rangle$.

Proof. (i) With Eq. (13) and Eq. (21), we have

$$\begin{aligned}&|\vec{a}\rangle\langle\vec{a}|+|\vec{b}\rangle\langle\vec{b}|\\ &=\left(\frac{g_k+\mathbb{1}}{2}+\frac{-g_k+\mathbb{1}}{2}\right)\prod_{\substack{i=1\\ i\neq k}}^{n}\frac{(-1)^{a_i}g_i+\mathbb{1}}{2}\\ &=\prod_{\substack{i=1\\ i\neq k}}^{n}\frac{(-1)^{a_i}g_i+\mathbb{1}}{2}\,.\end{aligned} \tag{23}$$

Since g_k cancels in Eq. (23), the explicit form of the generators g_i implies that, in this equation, there is no Y on qubit k. Since Y is the only Pauli matrix that changes under partial transposition, $|\vec{a}\rangle\langle\vec{a}|+Z_k|\vec{a}\rangle\langle\vec{a}|Z_k$ is invariant under T_k.

(ii) Using again Eqs. (13) and (21) yields

$$|\vec{a}\rangle\langle\vec{a}|+|\vec{b}\rangle\langle\vec{b}|=\left(\prod_{i\in\mathcal{N}(k)}\frac{(-1)^{a_i}g_i+\mathbb{1}}{2}+\prod_{i\in\mathcal{N}(k)}\frac{(-1)^{a_i+1}g_i+\mathbb{1}}{2}\right)\frac{(-1)^{a_k}g_k+\mathbb{1}}{2}\prod_{i\notin\tilde{\mathcal{N}}(k)}\frac{(-1)^{a_i}g_i+\mathbb{1}}{2}\,. \tag{24}$$

The expression in the first brackets can be simplified to

$$\begin{aligned}\prod_{i\in\mathcal{N}(k)}\frac{(-1)^{a_i}g_i+\mathbb{1}}{2}+\prod_{i\in\mathcal{N}(k)}\frac{(-1)^{a_i+1}g_i+\mathbb{1}}{2}&=2^{-N}\left(\sum_{\vec{x}\in\{0,1\}^N}(-1)^{\vec{a}\vec{x}}\prod_{i\in\mathcal{N}(k)}g_i^{x_i}+\sum_{\vec{x}\in\{0,1\}^N}(-1)^{\vec{a}\vec{x}+\sum_i x_i}\prod_{i\in\mathcal{N}(k)}g_i^{x_i}\right)\\ &=2^{-N+1}\left(\sum_{\text{even}}(-1)^{\vec{a}\vec{x}}\prod_{i\in\mathcal{N}(k)}g_i^{x_i}\right)\,,\end{aligned} \tag{25}$$

where we defined $N = |\mathcal{N}(k)|$ and $\sum_{\text{even}}$ is the sum over all $\vec{x} \in \{0,1\}^N$ for which $\sum_{i=1}^N x_i$ is even. Combining Eqs. (24) and (25) yields

$$|\vec{a}\rangle\langle\vec{a}| + |\vec{b}\rangle\langle\vec{b}| = 2^{-N}\left(\sum_{\text{even}}(-1)^{\vec{a}\vec{x}}\prod_{i\in\mathcal{N}(k)} g_i^{x_i}\right)(-1)^{a_k} g_k \prod_{i\notin\widetilde{\mathcal{N}}(k)} \frac{(-1)^{a_i}g_i + \mathbb{1}}{2}$$
$$+ 2^{-N}\left(\sum_{\text{even}}(-1)^{\vec{a}\vec{x}}\prod_{i\in\mathcal{N}(k)} g_i^{x_i}\right)\prod_{i\notin\widetilde{\mathcal{N}}(k)} \frac{(-1)^{a_i}g_i + \mathbb{1}}{2}\,. \tag{26}$$

In Eq. (26), the right-hand side is invariant under T_k. To see this, consider the first part of the sum, in which all terms contain g_k and an even number of generators in $\mathcal{N}(k)$. Together with the form of the generators [see Eq. (10)], this implies that there is no Y on qubit k.
The second part of the right-hand side of Eq. (26) does not have a Y on qubit k either, since it does not contain g_k. Consequently, Eq. (26) is invariant under T_k.

□

Let us now return to the main proof in which we will need Lemmas 1, 2 and 3.

Proposition 2. *Let $|Cl_n\rangle\langle Cl_n|$ be an n-qubit linear cluster state with $n > 3$ and g_i the generators of its stabilizer group. Let $\mathcal{B} = \{\beta_i\}$ be a subset of all qubits such that $\widetilde{\mathcal{N}}(\beta_i) \cap \widetilde{\mathcal{N}}(\beta_j) = \{\}$ for $i \neq j$. We define $m = |\mathcal{B}|$. Let $\sum_{\vec{s}}$ be the sum over all vectors $\vec{s}$ of length m with elements $s_i = \pm 1$ that contain at least two elements that are -1, i.e., $\sum_{i=1}^m s_i \leq m-4$. Then,*

$$W_{\text{Cln}} = \frac{1}{2}\mathbb{1} - |Cl_n\rangle\langle Cl_n| - \frac{1}{2}\left(\sum_{\vec{s}}\prod_{i\in\mathcal{B}}\frac{s_i g_i + \mathbb{1}}{2}\right) \tag{27}$$

is a fully decomposable witness for $|Cl_n\rangle\langle Cl_n|$, i.e., for any strict subset M of all qubits, it can be written in the form $W_{Cl} = P_M + Q_M^{T_M}$, where $P_M \geq 0, Q_M \geq 0$.

For the sake of brevity, we define $P_+ = \sum_{\vec{s}}\prod_{i\in\mathcal{B}}\frac{s_i g_i + \mathbb{1}}{2}$. Note that P_+ is a sum of all projectors onto graph state basis vectors that contain at least two excitations in $\mathcal{B}$, i.e., two bits that equal one. For example, if we choose $\mathcal{B} = \{1,4,7,\dots\}$ for the sake of illustration,

$$P_+ = \sum_{\vec{x}\in\{0,1\}^{n-m}} \big(|0x_1x_21x_3x_41\dots\rangle\langle 0x_1x_21x_3x_41\dots|$$
$$+ |1x_1x_20x_3x_41\dots\rangle\langle 1x_1x_20x_3x_41\dots|$$
$$+ |1x_1x_21x_3x_40\dots\rangle\langle 1x_1x_21x_3x_40\dots|$$
$$+ |1x_1x_21x_3x_41\dots\rangle\langle 1x_1x_21x_3x_41\dots|$$
$$+ \ \dots\big) \tag{28}$$

This also illustrates why the presented construction does not work for linear cluster states that consist of three or less qubits. In such a case, $\mathcal{B}$ could only have one element, since the elements' neighborhoods cannot overlap. Then, however, one cannot fulfill the condition that there must always be two bits in $\mathcal{B}$ that equal one. Note that sets $\mathcal{B} = \{1,4\}$ and $\mathcal{B} = \{1,4,7\}$ for four and seven qubits, respectively, result in the witnesses W_{Cl4} and W_{Cl7} of Eqs. (6) and (8) of the main text. Moreover, these choices of $\mathcal{B}$ lead to the noise tolerance of Eq. (9) of the main text.

Proof. First, we will provide a way to construct an appropriate operator $P_M \geq 0$ for every M. The main trick is to do this in such a way that this operator is invariant under certain partial transpositions. Second, we will use these P_M to show that $Q_M = (W - P_M)^{T_M} \geq 0$ for every M.
Note that we order the qubits β_i in a canonical way such that $\beta_i < \beta_{i+1}$. To simplify notation, we define bitstring $\vec{w}_i$ for every $\widetilde{\mathcal{N}}(\beta_i)$ (of $|\widetilde{\mathcal{N}}(\beta_i)|$ bits length) in such a way that its bits define to which partition the correspoding qubits belongs. $\vec{w}_i = (\omega_{-1}, \omega_0, \omega_{+1})$, $\omega_j \in \{0,1\}$, means that qubit $\beta_i + j$ does not belong to M if $\omega_j = 0$ and it belongs to M if $\omega_j = 1$. For β_i being the first qubit, we consider $\vec{w}_i = (\omega_0, \omega_{+1})$; for β_i being the last qubit, we use $\vec{w}_i = (\omega_{-1}, \omega_0)$.
Given M, we proceed in the following way to construct P_M.

1. Start with $P_M^{(0)} = |Cl_n\rangle\langle Cl_n| = |0\dots0\rangle\langle 0\dots0|$.

2. Loop through the values of $i = 1,2,\dots,m$ and update $P_M^{(i)}$ until you reach $i > m$. Then proceed with step 3. The update of $P_M^{(i)}$ is carried out in the following way.

   (i) If $|\widetilde{\mathcal{N}}(\beta_i)| = 2$:

      I If $\vec{w}_i = 01$ or $\vec{w}_i = 10$, then $P_M^{(i)} = P_M^{(i-1)} + Z_k P_M^{(i-1)} Z_k$, where $k \in \mathcal{N}(\beta_i)$.

      II If $\vec{w}_i = 11$ or $\vec{w}_i = 00$, let $P_M^{(i)} = P_M^{(i-1)}$.

   (ii) If $|\widetilde{\mathcal{N}}(\beta_i)| = 3$:

I If $\vec{w}_i = 110$ or $\vec{w}_i = 001$, then $P_M^{(i)} = P_M^{(i-1)} + Z_{\beta_i+1} P_M^{(i-1)} Z_{\beta_i+1}$.

II If $\vec{w}_i = 010$ or $\vec{w}_i = 101$, then $P_M^{(i)} = P_M^{(i-1)} + Z_{\beta_i-1} Z_{\beta_i+1} P_M^{(i-1)} Z_{\beta_i-1} Z_{\beta_i+1}$.

III If $\vec{w}_i = 100$ or $\vec{w}_i = 011$, then $P_M^{(i)} = P_M^{(i-1)} + Z_{\beta_i-1} P_M^{(i-1)} Z_{\beta_i-1}$.

IV In all other cases, let $P_M^{(i)} = P_M^{(i-1)}$.

3. Let r be the number of times that one of the cases 2.(i).I, 2.(ii).I, 2.(ii).II or 2.(ii).III occurred, i.e., the number of steps in which $P_M^{(i)}$ changed.
If $r \leq 1$, define

$$P_M = 0\,. \tag{29}$$

Let t be the value of i for which $P_M^{(i)}$ was changed the last time, i.e., $P_M^{(i)} = P_M^{(t)}\ \forall\, i > t$. If $r > 1$, define

$$P_M = P_M^{(t-1)} - |Cl_n\rangle\langle Cl_n|\,. \tag{30}$$

In other words, r equals the number of bits $\beta_i \in \mathcal{B}$ which do not lie at the border of M (or $\overline{M}$), i.e. they do not obey

$$\widetilde{\mathcal{N}}(\beta_i) \subseteq M \text{ or } \widetilde{\mathcal{N}}(\beta_i) \subseteq \overline{M}\,. \tag{31}$$

Thus, for $r = 0$, all $\beta_i \in \mathcal{B}$ obey this property, for $r = 1$, there is one exception.
Note that the operator P_M constructed via the given algorithm is either zero or a sum of one-dimensional projectors onto basis states, i.e.,

$$P_M = \sum_{\vec{a}} |\vec{a}\rangle\langle\vec{a}|\,. \tag{32}$$

This can be seen by the fact that $P_M^{(0)} = |Cl_n\rangle\langle Cl_n| = |0\ldots0\rangle\langle0\ldots0|$, the application of Z only flips a bit and finally $|Cl_n\rangle\langle Cl_n|$ is subtracted again. Let us illustrate the algorithm by a concrete example.
Example of the algorithm: Consider an eight-qubit cluster state and $M = \{2, 3, 5\}$. We choose $\mathcal{B} = \{1, 4, 7\}$. Then, the algorithm proceeds as follows.

- $P_M^{(0)} = |Cl_8\rangle\langle Cl_8| = |00000000\rangle\langle00000000|$

- Step $i = 1$, i.e., $\beta_1 = 1$: Since $\vec{w}_1 = 01$, case 2.(i).I applies and

$$\begin{aligned} P_M^{(1)} &= |00000000\rangle\langle00000000| \\ &\quad + Z_2|00000000\rangle\langle00000000|Z_2 \\ &= |00000000\rangle\langle00000000| \\ &\quad + |01000000\rangle\langle01000000|\,. \end{aligned} \tag{33}$$

- Step $i = 2$, i.e., $\beta_2 = 4$: Since $\vec{w}_2 = 101$, case 2.(ii).II applies and

$$\begin{aligned} P_M^{(2)} &= P_M^{(1)} + Z_3 Z_5 P_M^{(1)} Z_3 Z_5 \\ &= |00000000\rangle\langle00000000| \\ &\quad + |01000000\rangle\langle01000000| \\ &\quad + |00101000\rangle\langle00101000| \\ &\quad + |01101000\rangle\langle01101000|\,. \end{aligned} \tag{34}$$

- Step $i = 3$, i.e., $\beta_3 = 7$: We have $\vec{w}_3 = 000$. Thus, case 2.(ii).IV applies and $P_M^{(3)} = P_M^{(2)}$.

- As $i = 4 > m = 3$, we abort the loop. $P_M^{(i)}$ changed in two steps, i.e., $r = 2$. Moreover, since $P_M^{(i)}$ did not change in the last step, i.e., the third step, $t = 2$. Then,

$$\begin{aligned} P_M &= P_M^{(t-1)} - |Cl_7\rangle\langle Cl_7| \\ &= P_M^{(1)} - |00000000\rangle\langle00000000| \\ &= |01000000\rangle\langle01000000| \end{aligned} \tag{35}$$

Let us now return to the general case and understand the properties of the operator P_M for an arbitrary M. The construction uses Lemma 3 to ensure that, in every step, either

$$\left(P_M^{(i)}\right)^{T_{M_i}} = \left(P_M^{(i)}\right)^{T_{\widetilde{\mathcal{N}}(\beta_i)}} \tag{36a}$$

or

$$\left(P_M^{(i)}\right)^{T_{M_i}} = P_M^{(i)} \tag{36b}$$

hold, where we defined $M_k = M \cap \widetilde{\mathcal{N}}(\beta_k)$.
To see that Eqs. (36) hold, it is best to consider an example: Assume that $\vec{w}_i = 010$. Then, case 2.(ii).II of the algorithm applies. As seen above, $P_M^{(i-1)}$ has the form $P_M^{(i-1)} = \sum_{\vec{b}} |\vec{b}\rangle\langle\vec{b}|$. Thus,

$$P_M^{(i)} = \sum_{\vec{b}} |\vec{b}\rangle\langle\vec{b}| + Z_{i-1}Z_{i+1}|\vec{b}\rangle\langle\vec{b}|Z_{i+1}Z_{i-1}\,. \tag{37}$$

Then, Lemma 2.(ii) implies that Eq. (36b) holds.
Eqs. (36) hold in every step, i.e., for $i = j$ and for $i = k$, where $j \neq k$. According to the premise of non-overlapping neighborhoods of the qubits in $\mathcal{B}$, we have $\widetilde{\mathcal{N}}(\beta_j) \cap \widetilde{\mathcal{N}}(\beta_k) = \{\}$. Therefore, the partial transpositions in Eqs. (36) for $i = j$ always affect qubits different from the ones that are affected by the partial transpositions for $i = k$. For this reason, Eqs. (36) for $P_M^{(t-1)}$ hold with respect to to every value of k, except for $k = t$. More precisely,

$$\left(P_M^{(t-1)}\right)^{T_{M_k}} = \left(P_M^{(t-1)}\right)^{T_{\widetilde{\mathcal{N}}(\beta_k)}} \tag{38a}$$

or

$$\left(P_M^{(t-1)}\right)^{T_{M_k}} = P_M^{(t-1)} \tag{38b}$$

is true for every $k \neq t$. We will use this important property later.

Let us proceed with the proof. Since P_M is zero or has the form of Eq. (32), we know that $P_M \geq 0$. Thus, it remains to show that $(W_{Cl} - P_M)^{T_M} = Q_M \geq 0,\ \forall\, M$.

Operators that are diagonal in a graph state basis can be written in the form $\sum_{\vec{x}} \prod_{i=1}^n g_i^{x_i}$, where the sum runs over binary strings $\vec{x}$ that depend on the operator. Since any partial transposition can at most introduce minus signs in some terms of this sum, such operators remain diagonal under any partial transposition. Therefore, it is enough to prove that

$$\langle\vec{k}|\,(W_{Cl} - P_M)^{T_M}\,|\vec{k}\rangle \geq 0,\ \forall\, M, |\vec{k}\rangle\,. \tag{39}$$

Note that $P_+ = \sum_{\vec{s}} \prod_{i\in\mathcal{B}} \frac{s_i g_i + \mathbb{1}}{2}$ is invariant under any partial transposition, since it does not contain any Y operators due to the form of the generators and the fact that no two qubits in $\mathcal{B}$ have a common neighbor. Therefore, together with the explicit form of the witness given in Eq. (27), we can rewrite Eq. (39) as

$$\frac{1}{2} - \frac{1}{2}\langle\vec{k}|P_+|\vec{k}\rangle - \langle\vec{k}|\,(|Cl_n\rangle\langle Cl_n| + P_M)^{T_M}\,|\vec{k}\rangle \geq 0,\ \forall M, |\vec{k}\rangle. \tag{40}$$

In order to prove this, we distinguish two different cases:

1. $\boxed{\langle\vec{k}|P_+|\vec{k}\rangle \neq 0 \Leftrightarrow \langle\vec{k}|P_+|\vec{k}\rangle = 1}$

   Note that the equivalence can best be see in Eq. (28). Also, Eq. (28) implies that there must be at least two bits of $\vec{k}$, say $k_{i_0}, k_{j_0} \in \mathcal{B}$, with $i_0 \neq j_0$, such that $k_{i_0} = k_{j_0} = 1$.

   In the case $P_M = 0$, Eq. (40) and $\langle\vec{k}|P_+|\vec{k}\rangle = 1$ are equivalent to

   $$-\langle\vec{k}|\,(|Cl_n\rangle\langle Cl_n|)^{T_M}\,|\vec{k}\rangle \geq 0\,\forall\, M, |\vec{k}\rangle\,. \tag{41}$$

   To see that the left-hand side always vanishes, one uses that $P_M = 0$ is equivalent to $r \leq 1$, i.e., cases 2.(i).I or 2.(ii).I or 2.(ii).II or 2.(ii).III in the algorithm occurred at most once. This means that $\widetilde{\mathcal{N}}(\beta_i) \subseteq M$ or $\widetilde{\mathcal{N}}(\beta_i) \subseteq \overline{M}$ holds for all qubits $\beta_i \in \mathcal{B}$ with at most one exception, namely β_t. With $k_{i_0} = k_{j_0} = 1$, Lemma 1 can be applied to see that Eq. (41) vanishes.

   In the case $P_M \neq 0$, Eq. (40) can be simplified using $\langle\vec{k}|P_+|\vec{k}\rangle = 1$ to

   $$\begin{aligned} & -\langle\vec{k}|\,(|Cl_n\rangle\langle Cl_n| + P_M)^{T_M}\,|\vec{k}\rangle \geq 0\,\forall\, M, |\vec{k}\rangle \\ \Leftrightarrow\ & -\langle\vec{k}|\left(P_M^{(t-1)}\right)^{T_M}|\vec{k}\rangle \geq 0\,\forall\, M, |\vec{k}\rangle\,. \end{aligned} \tag{42}$$

   Here, the definition of P_M, Eq. (30), has been used. Now, P_M and therefore $P_M^{(t-1)}$ consists of a sum of projectors onto graph basis states $|\vec{a}\rangle$ (see Eq. (32)). Since the algorithm starts with $P_M^{(0)} = |0\ldots0\rangle\langle0\ldots0|$ and never flips any bits on the qubits $\beta_i \in \mathcal{B}$, these states $|\vec{a}\rangle$ obey $a_{\beta_i} = 0,\ \forall\, i = 1,\ldots,m$. Also, depending on whether $i_0 = t$ or $j_0 = t$, Eqs. (38) can be applied to whichever of these two qubits is different from t. The invariance given by Eqs. (38) then implies that this qubit can be treated as if it lay on the border of M (or $\overline{M}$) and therefore Lemma 1 yields

   $$\langle\vec{k}|\left(P_M^{(t-1)}\right)^{T_M}|\vec{k}\rangle = 0 \tag{43}$$

   and therefore Eq. (42) holds.

2. $\boxed{\langle\vec{k}|P_+|\vec{k}\rangle = 0}$

To show that Eq. (40) holds, we need to prove that

$$\langle\vec{k}|\,(|Cl_n\rangle\langle Cl_n| + P_M)^{T_M}\,|\vec{k}\rangle \leq \frac{1}{2}\,. \tag{44}$$

In the case $P_M \neq 0$, P_M is given by Eq. (32) and Eq. (44) is equivalent to

$$\langle\vec{k}|\left(|Cl_n\rangle\langle Cl_n| + \sum_{\vec{a}} |\vec{a}\rangle\langle\vec{a}|\right)^{T_M}|\vec{k}\rangle \leq \frac{1}{2}\,. \tag{45}$$

Note that $|Cl_n\rangle\langle Cl_n| + \sum_{\vec{a}} |\vec{a}\rangle\langle\vec{a}| = P_M^{(t-1)}$ consists of 2^{r-1} terms, as one starts with one term and doubles this number $(r-1)$ times to obtain $P_M^{(t-1)}$. Therefore, it is enough to show that

$$\langle\vec{k}|\,(|Cl_n\rangle\langle Cl_n|)^{T_M}\,|\vec{k}\rangle \leq 2^{-r} \tag{46}$$

and

$$\langle\vec{k}|\,(|\vec{a}\rangle\langle\vec{a}|)^{T_M}\,|\vec{k}\rangle \leq 2^{-r}\,\forall\, |\vec{a}\rangle \tag{47}$$

holds. We will do this by using Lemma 2. However, since the vectors $|\vec{a}\rangle$ are basis vectors of the graph state basis of $|Cl_n\rangle$, $|\vec{a}\rangle$ and $|Cl_n\rangle$ are LU-equivalent. Therefore, they have the same Schmidt coefficients and Lemma 2 results in the same upper bounds. For this reason, it suffices to show only one of these upper bounds, namely

$$\langle\vec{k}|\,(|Cl_n\rangle\langle Cl_n|)^{T_M}\,|\vec{k}\rangle \leq 2^{-r}\,. \tag{48}$$

The idea behind the subsequent proof of Eq. (48) is that, every time $P_M^{(i)}$ changes in our algorithm, r is increased, but at the same time, a Bell pair which has Schmidt coefficients $\{\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\}$ is created which results in a smaller upper bound according

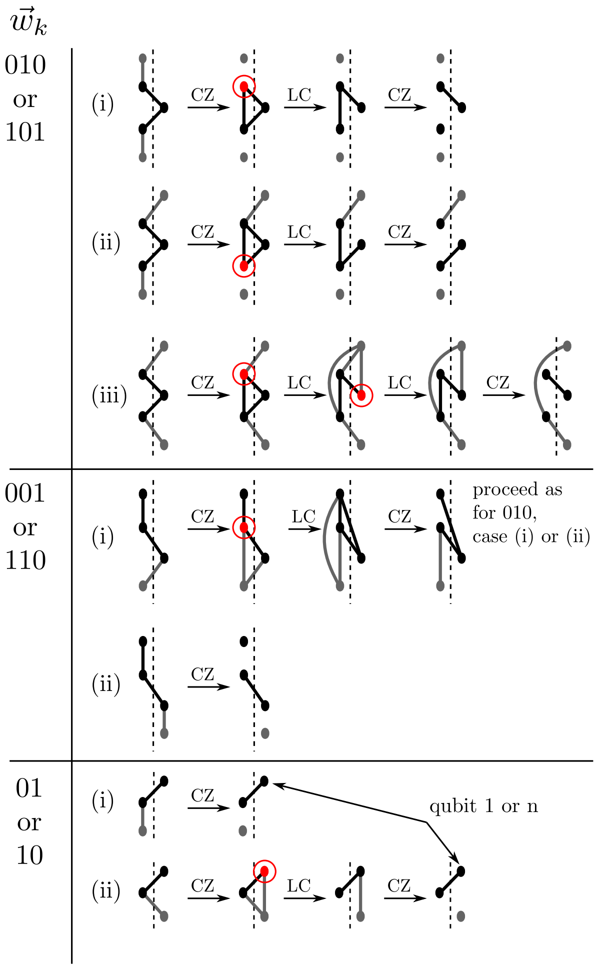

FIG. 1: For every case in which $P_M^{(i)}$ changes in the algorithm, we explicitly demonstrate that it is possible to create a Bell pair via a sequence of local transformations. Note that the cases $\vec{w}_k = 100$ and $\vec{w}_k = 011$ follow from $\vec{w}_k = 110$ or $\vec{w}_k = 001$ due to symmetry reasons.

to Lemma 2.
To easily calculate the Schmidt coefficients of $|Cl_n\rangle$ with respect to bipartition $M|\overline{M}$, we transform $|Cl_n\rangle$ via local transformations into disconnected Bell pairs whose qubits are in different partitions. In order not to change the Schmidt coefficients during this transformation, we apply the following two operations to the graph.

- *Local Complementation (LC)*: Pick a qubit j and invert its neighborhood graph. In other words, if between two neighbors of j there is an edge, delete it. If between two neighbors of j there is no edge, add an edge between them. Since this operation corresponds to the application of local unitaries, the Schmidt coefficients remain unchanged.

- *Controlled-Z (CZ)*: Apply a controlled-Z operation to two qubits in the same partition. If they are connected by an edge, this will delete the edge. If they are not connected, this will create an edge. As this unitary operation is only applied on qubits in the same partition, it does not change the Schmidt coefficients.

Since these transformations can be applied to each set $\widetilde{\mathcal{N}}(\beta_j)$ individually to create a Bell pair between the two partitions (see Fig. 1) it is possible to end up with distinct Bell pairs between the two partitions.
Moreover, the number q of Bell pairs that one can create by transforming the qubits in each $\widetilde{\mathcal{N}}(\beta_j)$ separately obeys $q \geq r$, as every time that $P_M^{(i)}$ is changed in the algorithm (including step $i = t$) corresponds to a configuration shown in Fig. 1. Furthermore, each of these configurations allow for the creation of a disconnected Bell pair.
Since the Schmidt coefficients of a single Bell pair are $\{\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\}$, Lemma 2 yields

$$\langle\vec{k}|\left(|Cl_n\rangle\langle Cl_n|\right)^{T_M}|\vec{k}\rangle \leq 2^{-q}\,, \tag{49}$$

With $q \geq r$, this implies that Eq. (48) holds.
In the case $P_M = 0$, we need to show that

$$\langle\vec{k}|\left(|Cl_n\rangle\langle Cl_n|\right)^{T_M}|\vec{k}\rangle \leq \frac{1}{2}\,. \tag{50}$$

Note that $P_M = 0$ is equivalent to $r \leq 1$. On one hand, $r = 0$ means that for each $i = 1, \ldots, n$, either $\widetilde{\mathcal{N}}(i) \subseteq M$ or $\widetilde{\mathcal{N}}(i) \subseteq \overline{M}$ holds. Since $M \neq \{1, \ldots, n\}$, deleting all edges within M and $\overline{M}$ will lead to at least one Bell pair, which has Schmidt coefficients $\{\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\}$. With Lemma 2, Eq. (50) holds.
On the other hand, for $r = 1$, Eq. (50) also holds, since due to Fig. 1 the Schmidt coefficients are given by at least one Bell pair. Therefore, the maximal coefficient is $\frac{1}{2}$.
This finishes the proof of Proposition 2.

□

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